

GRPO Reward for NGSC: Suggested Forms

Y_c : ground-truth mask of class c ; $\hat{M}_{a,c}$: binary mask for class c from action a ; Ω : image domain; $\mathcal{C}_{\text{tar}}$: target classes; group $a_1, \dots, a_G$ sampled per image

Component	Suggested form	Why (issue it fixes)
Per-class reward, present ($Y_c \neq \emptyset$)	$R_c(a) = w_D \text{Dice}(\hat{M}_{a,c}, Y_c) + w_L \text{F1}_{\text{les}}(\hat{M}_{a,c}, Y_c) + w_B \text{BF1}(\hat{M}_{a,c}, Y_c),$ $w_D + w_L + w_B = 1$ (e.g. 0.6/0.3/0.1)	Dice alone barely moves on small lesions and ties across similar actions; component-level F1_{les} restores signal exactly where seeds fail (KiTS), BF1 keeps boundary quality rewarded.
Per-class reward, absent ($Y_c = \emptyset$)	$R_c(a) = \mathbf{1}[\hat{M}_{a,c} = \emptyset] + (1 - \mathbf{1}[\hat{M}_{a,c} = \emptyset]) \left(1 - \frac{ \hat{M}_{a,c} }{ \Omega }\right) (1 - m),$ margin $m \approx 0.2$	$1 - \hat{M} / \Omega $ is too flat: a small false positive scores ≈ 0.99 vs. 1.0 for abstaining. The margin separates “empty” from “any non-empty” so the gate δ_c actually learns.
Multi-class aggregation	$R(a) = \frac{\sum_{c \in \mathcal{C}_{\text{tar}}} \omega_c R_c(a)}{\sum_c \omega_c}, \quad \omega_c = 1 \text{ if } Y_c \neq \emptyset, \text{ else } \omega_0 \approx \frac{1}{ \mathcal{C}_{\text{tar}} }$	Figure defines only the single-class case. Each class follows its own branch; present classes dominate, absent classes still teach abstention without flooding the sum.
Advantage	$\hat{A}_g = \frac{R(a_g) - \bar{R}}{\text{std}(R_{1:G}) + \varepsilon} \cdot \mathbf{1}[\text{std}(R_{1:G}) > \varepsilon_{\min}];$ alternative: RLOO $\hat{A}_g = R_g - \frac{1}{G-1} \sum_{g' \neq g} R_{g'}$	Group strictly per image, never across a batch (branch statistics differ). Skip zero-variance groups: all-abstain normals or identical masks otherwise yield $0/\varepsilon$ noise. RLOO needs no skip rule.
Objective	$\mathcal{J}(\theta) = \frac{1}{G} \sum_g \min(\rho_g \hat{A}_g, \text{clip}(\rho_g, 1 \pm \epsilon) \hat{A}_g) - \beta D_{\text{KL}}(\pi_\theta \ \pi_{\text{ref}}) - \zeta \widehat{\text{miss}}, \quad \widehat{\text{miss}} = \Pr[\hat{M} = \emptyset \mid y_{\text{pres}}=1]$	Last term (small $\zeta \approx 0.1$, batch estimate on calibration positives) blocks the degenerate optimum where abstaining everywhere is safest; equivalently monitor miss rate and early-stop.
Group sampling	$a_1, \dots, a_G$ distinct configurations (sample grids without replacement); seed factor provides within-group diversity	Adjacent η grid steps often give identical $\hat{M}$ and tied R at 14×14 patch granularity; distinct configs avoid wasted zero-advantage samples.

Conventions kept from the original figure: exactly one branch per class per sample, rewards never added across branches, $0 \leq R \leq 1$ (regularizers live in $\mathcal{J}$, not in R).